Straight Line Fitting: A Bayesian Solution

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1991

Fitting the "best" straight line to a scatter plot of data $D = \{(x_1, y_1)...(x_n, y_n)\}$ in which both variables x_i, y_i are subject to unknown error is undoubtedly the most common problem of inference faced by scientists, engineers, medical researchers, and economists. The problem is to estimate the parameters α, β in the straight line equation $y = \alpha + \beta x$, and assess the accuracy of the estimates. Whenever we try to discover or estimate a relationship between two factors we are almost sure to be in this situation. But from the viewpoint of orthodox statistics the problem turned out to be a horrendous can of worms; generations of efforts led only to a long line of false starts, and no satisfactory solution. We give the Bayesian solution to the problem, which turns out to be eminently satisfactory and straightforward, although a little tricky in the derivation. However, not much of the final result is really new. Arnold Zellner (1971) gave a very similar solution long ago, but it went unnoticed by those who had the most need to know about it. We give a pedagogical introduction to the problem and add a few final touches, dealing with choice of priors and parameterizations. In any event, whether or not the following solution has anything new in it, the currently great and universal importance of the problem would warrant bringing the result to the attention of the scientific community. Many workers, from astronomers to biologists, are still struggling with the problem, unaware that the solution is known.

INTRODUCTION

The following discussion is an instructive case history showing how nontrivial Bayesian results evolve. It illustrates three very important points:

In estimating α, β, we have potentially $n+2$ nuisance parameters, $\{X_1 \ldots X_n, \sigma_x, \sigma_y\}$. Denoting prior information by I, the most general solution will then have a joint prior pdf for $n+4$ parameters:

$$p(\alpha, \beta, X_1 \ldots X_n, \sigma_x, \sigma_y | I) = p(\alpha, \beta | I) p(X_1 \ldots X_n, \sigma_x, \sigma_y | \alpha, \beta, I) \qquad (1)$$

and an unending variety of different kinds of prior information I might be expressed by this function, which we understand is to be properly normalized to unit integral. By the product rule, $p(\alpha, \beta | I)$ can always be factored as shown, although the possibility of a Borel-Kolmogorov paradox should be kept in mind. That is, by a probability $p(A|\alpha, \beta, I)$ conditional on point values of α, β, we must understand the limit of the well-defined

$$P(A|d\alpha d\beta I) = \frac{P(A, d\alpha d\beta | I)}{P(d\alpha d\beta | I)} \qquad (2)$$

as $d\alpha \to 0, d\beta \to 0$. To avoid ambiguity it is necessary to prescribe the exact way in which the limit is to be approached.

For example, if we set $d\alpha = \epsilon g(\beta)$ and pass first to the limit $\epsilon \to 0$, our final results will in general depend on which function $g(\beta)$ we chose. But there is no 'right choice' or 'wrong choice' because it is for us to say which limit we want to take; i.e., which problem we want to solve. Any choice presumably corresponds to a legitimate problem that we might want to reason about, and probability theory will then give us the correct solution to that problem. But having made one choice, we must stick to that choice throughout the calculation, otherwise we are switching problems in midstream and are pretty sure to generate contradictions.

This is Sermon #1 on mathematical limits; although it was given long ago by Kolmogorov, many who try to do probability calculations still fail to heed it and get themselves into trouble. The moral is that, unless they are defined in the statement of a problem, probabilities of the form $p(A|\ldots I)$ conditional on point values of a parameter, have no meaning until the specific limiting process is stated. More generally, probabilities conditional on any propositions of probability zero, are undefined.

In the following we use the abbreviations

$$x = \{x_1 \ldots x_n\}, \quad Y = \{Y_1 \ldots Y_n\}, \quad \text{etc.}, \tag{3}$$

so that our data are denoted by $D = x, y$. Then the most general sampling pdf would have the functional form

$$p(x, y|\alpha, \beta, X, \sigma_x, \sigma_y, I) \tag{4}$$

and the most general solution we contemplate here would have the form

$$p(\alpha, \beta|x, y, I) = p(\alpha, \beta|I)\frac{p(x, y|\alpha, \beta, I)}{p(x, y|I)} \tag{5}$$

in which

$$p(x, y|\alpha, \beta, I) = \int d^n X d\sigma_x d\sigma_y \, p(x, y|\alpha, \beta, X, \sigma_x, \sigma_y, I) p(X, \sigma_x, \sigma_y|\alpha, \beta, I) \tag{6}$$

$$p(x, y|I) = \int d\alpha d\beta \, p(x, y|\alpha, \beta, I) p(\alpha, \beta|I) \tag{7}$$

and, writing $x^*, y^* = \{(x_{m+1}, y_{m+1}) \ldots (x_{m+n}, y_{m+n})\}$, our most general predictive distribution is

$$p(y^*|x^*, x, y, I) = \int p(y^*|x^*, \alpha, \beta, I) p(\alpha, \beta|x, y, I) d\alpha d\beta \tag{8}$$

This defines our present horizon (but having found this solution, its extension to such details as more than two variables, correlated noise, noise known to vary with x, etc. is an easy homework problem, involving little more than promoting some of our symbols from numbers to matrices).

SPECIAL CASES

Note first how the standard solutions are contained in this as special cases. If, as is almost universally supposed, the prior pdf for the errors factors completely:

$$p(e,f|I) = \prod_{i=1}^{n} p(e_i|I)p(f_i|I) \tag{9}$$

with common Gaussian distributions

$$p(e_i|I) = \frac{1}{\sqrt{2\pi}\sigma_x}\exp\left(-\frac{e_i^2}{2\sigma_x^2}\right), \quad p(f_i|I) = \frac{1}{\sqrt{2\pi}\sigma_y}\exp\left(-\frac{f_i^2}{2\sigma_y^2}\right) \tag{10}$$

Now, dropping the prior information symbol I, which we suppose henceforth to be hidden in the right-hand side of all our probabilities, our sampling pdf is

$$p(x,y|\alpha,\beta,\sigma_x,\sigma_y,X) = \prod_{i=1}^{n}\frac{1}{2\pi\sigma_x\sigma_y}\exp\left\{-\frac{(y_i-\alpha-\beta X_i)^2}{2\sigma_y^2} - \frac{(x_i-X_i)^2}{2\sigma_x^2}\right\} \tag{11}$$

which we note factors as shown. By Bayes' theorem,

$$p(\alpha,\beta|x,y,\sigma_x,\sigma_y) = \int d^nX p(\alpha,\beta,X|x,y,\sigma_x,\sigma_y) = \int d^nX p(\alpha,\beta,X|\sigma_x,\sigma_y)\frac{p(x,y|\alpha,\beta,\sigma_x,\sigma_y,X)}{p(x,y|\sigma_x,\sigma_y)} \tag{12}$$

and whether σ_x, σ_y are known or unkwown, the solutions will depend on the data only through their first and second moments, which are sufficient statistics for α, β. Introducing the standard notations for the observed sample first and second moments,

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \quad \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i, \tag{13}$$

$$\overline{x^2} = \frac{1}{n}\sum x_i^2, \quad \overline{xy} = \frac{1}{n}\sum x_i y_i, \quad \overline{y^2} = \frac{1}{n}\sum y_i^2, \tag{14}$$

and the sample central moments and correlation coefficient

$$s_{xx} = \overline{x^2} - \bar{x}^2, \quad s_{xy} = \overline{xy} - \bar{x}\bar{y}, \quad s_{yy} = \overline{y^2} - \bar{y}^2, \quad r = \frac{s_{xy}}{\sqrt{s_{xx}s_{yy}}}, \tag{15}$$

we note for later purposes that eventually all solutions involve the fundamental quadratic form determined by the data

$$Q(\alpha,\beta) \equiv \frac{1}{n}\sum_{i=1}^{n}(y_i-\alpha-\beta x_i)^2 = C_{11}(\alpha-\hat{\alpha})^2 + 2C_{12}(\alpha-\hat{\alpha})(\beta-\hat{\beta}) + C_{22}(\beta-\hat{\beta})^2 + C_0 \tag{16}$$

where the matrix C is

$$C = \begin{pmatrix} 1 & \bar{x} \\ \bar{x} & \overline{x^2} \end{pmatrix} \tag{17}$$

and the requirement that (16) be an identity in α, β uniquely determines the coefficients:

$$\hat{\beta} = s_{xy}/s_{xx}$$

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x} \tag{18}$$

$$C_0 = s_{yy} - s_{xy}^2/s_{xx} = s_{yy}(1 - r^2).$$

Of these, $\hat{\alpha}$ and $\hat{\beta}$ are just the original least-squares estimates of α, β that one would make if the errors were only in the y_i-measurmenents. But before reaching the form $Q(\alpha, \beta)$, we have to integrate out the nuisance parameters $X_1 \ldots X_n$, and perhaps also σ_x, σ_y.

THE ONE-POINT PROBLEM

To orient our thinking about this, consider first the 'baby' problem of estimating X_1 given only the datum x_1. If σ_x^2 were known and we had only the data component x_1, from (11) we would have immediately for the posterior pdf for X_1:

$$p(X_1|x_1, I) = Ap(X_1|I) \exp\left[-\frac{(x_1 - X_1)^2}{2\sigma_x^2}\right] \tag{19}$$

where here and what follows, A always stands for a normalizing constant, not necessarily the same in all equations. Suppose our prior information had led us to estimate X_1 as about $x_0 \pm \delta$; we could indicate this by the prior pdf

$$p(X_1|I) = \frac{1}{\sqrt{2\pi}\delta^2} \exp\left[-\frac{(X_1 - x_0)^2}{2\delta^2}\right] \tag{20}$$

But we note that

$$\frac{(X_1 - x_0)^2}{\delta^2} + \frac{(x_1 - X_1)^2}{\sigma_x^2} = \frac{(X_1 - \bar{X}_1)^2}{\sigma^2} + (\text{const.}) \tag{21}$$

where the (const.) is independent of X_1, and

$$\bar{X}_1 = \frac{x_0/\delta^2 + x_1/\sigma_x^2}{1/\delta^2 + 1/\sigma_x^2} \tag{22}$$

$$\frac{1}{\sigma^2} = \frac{1}{\delta^2} + \frac{1}{\sigma_x^2} \tag{23}$$

whereupon (19) becomes

$$p(X_1|x_1, I) = \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left[-\frac{(X_1 - \bar{X}_1)^2}{2\sigma^2}\right] \tag{24}$$

We would estimate X_1 as (mean $\pm$ standard deviation)

$$(X_1)_{\text{est}} = \bar{X}_1 \pm \sigma \tag{25}$$

a weighted average of the prior estimate x_0 and the datum x_1, weighted according to the respective variances.

If we had almost no prior information about the unknown true value X_1, then $\delta \gg \sigma_x$ and this would reduce for all practical purposes to $(X_1)_{\text{est}} = x_1 \pm \sigma_x$. For example, if $\delta > 10\sigma_x$, then the exact solution is within one percent of this limiting value, and the prior information would hardly help at all. But if we had prior information fixing X_i to an uncertainty comparable to σ_x, this would evidently be cogent, enabling us to improve the accuracy of our estimates of α, β. This discussion of the 'baby' problem is to condition us to the usual argument for passage to an improper prior, $\delta \to \infty$. Usually, the data will be highly informative compared to our prior information (indeed, data which tell us little that we did not know already, would be hardly worth gathering). But if as usual our prior information is vague compared to the accuracy of the data, then whether we keep the prior with finite δ or pass to the limit of an improper prior, $\delta \to \infty$, makes no difference in the results. This is the conventional argument, surely valid for the simple problem being considered.

THE REAL PROBLEM

But note that the above passage to the limit is, in principle, to be carried out only at the end of the calculation. In the real problem, the properly normalized posterior distribution (5) is a ratio of two integrals, (6) and (7), and if we want to approach an improper prior it is the limit of the ratio, not the ratio of the limits, that should be carried out according to the rules of probability theory. The former limit is the well-behaved solution that we want; the latter may or may not exist. Depending on how rapidly the likelihood factor cuts off away from its peak, the separate integrals (6), (7) may diverge in the limit.

In the integrations over X_i it does not matter, because the Gaussian factors guarantee convergence of the integrals. Then we may behave in a rather reckless way and still get the right answer; but this is a rather unfortunate accident, that encourages bad mathematical habits that will fail on other similar-looking problems. For example, had we used a Cauchy noise distribution $p(e_i) \propto (a^2 + e_i^2)^{-1}$ instead of the Gaussian (10), the limit of the ratio would still be a perfectly well-behaved quantity, but the ratio of the limits would not exist. Our present problem involves not only the safe integration over the X_i, but also integration over σ_x and σ_y, for which the limit of the ratio continues to be well-behaved, but attempting to calculate instead the ratio of the limits can get us into trouble.

Admittedly, the point we are making is quite trivial, since if one does not see the distinction between the limit of a ratio and a ratio of the limits, he cannot even grasp the concept of a derivative dy/dx.

Nevertheless, the recent literature of probability theory has examples where the use of improper priors as limits of proper priors is rejected, because well-known authors failed to perceive this trivial point and tried to calculate the ratio of the limits instead of the limit of the ratio. See, for example, our exchange with DSZ over the Marginalization Paradox (Zellner, 1980).

Let us see how easy it is for the unwary to commit this error; but also how easy it is to avoid once we understand the point. If we assign uniform priors to the X_i on the above grounds, and the Jeffreys priors $d\sigma/\sigma$ to σ_x, σ_y, we may as noted integrate $X_1 \ldots X_n$ out of (11) without disaster, and this constructs for us the quadratic form $Q(\alpha, \beta)$:

$$p(\alpha, \beta, \sigma_x, \sigma_y | x, y, I) = \frac{A}{(\sigma_x \sigma_y)(\sigma_y^2 + \beta^2 \sigma_x^2)^{n/2}} \exp\left\{ -\frac{nQ(\alpha, \beta)}{2(\sigma_y^2 + \beta^2 \sigma_x^2)} \right\} \quad (26)$$

and now we face the mathematical subtlety that is the real point of all this discussion. If we try to get $p(\alpha, \beta | x, y, I)$ by integrating out σ_x, σ_y from this, the result diverges due to the factor $(\sigma_x \sigma_y)^{-1}$, which expresses the Jeffreys prior indicating ignorance of σ_x, σ_y.

But in (26) we have violated the rules of probablity theory by passing to the limit of improper priors before doing the normalizing integral; we are in effect trying to calculate the ratio of the limits. We got away with this with the X_i, but not with the σ_x, σ_y. Had we calculated the normalizing integral first for proper priors there could have been no divergence; then passing to the limit of the improper priors afterward would be a perfectly safe, uneventful procedure leading to the useful result that we want.

In 1965, the writer did not yet perceive this and was carried along by the arguments of Deming and Mandel, that the problem was indeterminate; this led to a long comedy of errors which I have seen others repeating many times since. My notebook entry of that time says: "This is symptomatic of the fact that the data of the problem do not provide any information at all about whether the errors are in x or in the y measurement. Then supposing Deming's parameter $\lambda = \sigma_y / \sigma_x$ known, we can see whether this enables us to get a Bayesian solution; instead of writing the joint prior proportional to $1/(\sigma_x \sigma_y)$, we use

$$p(\sigma_x, \sigma_y | I) \propto \frac{\delta(\sigma_y - \lambda \sigma_x)}{\sigma_x} \quad (27)$$

then integration over σ_y merely makes the substitution $\sigma_y = \lambda \sigma_x$, and it reduces to a convergent integral:

$$p(\alpha, \beta | x, y, I) = \frac{A}{(\lambda^2 + \beta^2)^{n/2}} \int_0^\infty \frac{d\sigma_x}{\sigma_x^{n+1}} \exp\left\{ -\frac{nQ(\alpha, \beta)}{2(\lambda^2 + \beta^2)} \right\} = \frac{A}{Q(\alpha, \beta)^{n/2}} \quad (28)$$

which is just the bivariate t-distribution that we would have had for the simpler regression problem in which the errors are only in the y-measurement and σ_y is completely unknown. Then, for example, we can integrate out α to get the posterior marginal pdf for β:

$$p(\beta | x, y, I) \propto [\gamma^2 + (\beta - \hat{\beta})^2]^{-(n-1)/2} \quad (29)$$

where $\gamma^2 = s_{yy}(1 - r^2)/s_{xx}$. As the initial pleasure at this nice result wore off, a little warning bell started ringing in my mind as it dawned on me that, unlike Deming's least squares result, (29) is independent of λ. How can it be that the

problem is indeterminate if λ is unknown; yet the solution when λ is known does not depend on λ?

A few years later, the answer suddenly seemed intuitively obvious. Instead of supposing λ known, make the change of variables $(\sigma_x, \sigma_y) \to (\sigma, \lambda)$ in (26):

$$\sigma = \sqrt{\sigma_y^2 + \beta^2 \sigma_x^2}, \quad \lambda = \sigma_y / \sigma_x \tag{30}$$

The jacobian is

$$\frac{\partial(\sigma_x, \sigma_y)}{\partial(\sigma, \lambda)} = \frac{\sigma}{\lambda^2 + \beta^2} \tag{31}$$

from which we find that the element of prior probability transforms as

$$\frac{d\sigma_x d\sigma_y}{\sigma_x \sigma_y} = \frac{d\lambda d\sigma}{\lambda \sigma} \tag{32}$$

and (26) becomes

$$p(\alpha, \beta, \sigma, \lambda | x, y, I) = A \cdot \frac{d\lambda}{\lambda} \frac{d\sigma}{\sigma^{n+1}} \exp\left\{ -\frac{nQ(\alpha, \beta)}{2\sigma^2} \right\} \tag{33}$$

At the time, I drew the conclusion that λ is completely decoupled from the problem:

$$p(\alpha, \beta, \sigma, \lambda | x, y, I) = p(\lambda | x, y, I) p(\alpha, \beta, \sigma | x, y, I) \tag{34}$$

so whatever prior we had assigned to λ would just integrate out again into a normalization constant, and contribute nothing to the final result. The algebra now seems to tell us that, far from being essential to make a determinate problem, λ is completely irrelevant to our problem! At least, from (33) it is clear how it can be that integrating out λ leads to divergence; yet supposing λ known leads to a result independent of λ.

The 'solution' which I offered at the 1973 Waterloo, Ontario meeting (Jaynes, 1976) is then

$$p(\alpha, \beta | x, y, I) = \int_0^\infty p(\alpha, \beta, \sigma | I) d\sigma \propto Q(\alpha, \beta)^{-n/2}, \tag{35}$$

the same as (28). But as the pleasure at this nice result wore off for the second time, it dawned on me that λ ought to be relevant to the problem after all. The result (35) is identical with what everybody, from Gauss on, had found for the case that σ_x is known to be zero; the measurement errors are only in y. In the opposite extreme, where the errors are only in x, the roles of x and y ought to be interchanged; but the quadratic form $Q(\alpha, \beta)$ is not a symmetric function of x_i and y_i. So where did I go wrong?

Let us go back to (33) and the conclusion we drew from it. Seeing only (33), it appears that not only is λ irrelevant to α, β, the data x, y tell us nothing about λ, for $p(\lambda | x, y, I) = p(\lambda | I)$. That is what we meant by saying that λ is completely decoupled from the problem.

The error in this reasoning was that (33) was derived only in the case of the Jeffreys prior (32); it has been shown thus far only that λ is decoupled for that prior. It turns out that exactly the same error of interpretation generated the "marginalization paradox" that was about to burst upon us (Dawid, et al, 1973; Jaynes, 1980). But for a general prior $f(\sigma_x, \sigma_y)d\sigma_x d\sigma_y$ the transformation would be, in place of (31),

$$f(\sigma_x, \sigma_y)d\sigma_x d\sigma_y = f(\sigma_x, \sigma_y)\frac{\sigma}{\lambda^2 + \beta^2}d\lambda d\sigma \tag{36}$$

Now the joint prior for λ and σ is

$$g(\lambda, \sigma) = \frac{\sigma}{\lambda^2 + \beta^2} f\left(\frac{\sigma}{\sqrt{\lambda^2 + \beta^2}}, \frac{\sigma\lambda}{\sqrt{\lambda^2 + \beta^2}}\right) \tag{37}$$

which is in general very far from being decoupled!

But there is still another error in what we have done, which Steve Gull recognized. We integrated out the X_i with respect to independent uniform priors on the grounds that our prior uncertainty δ is large compared to σ_x so its exact value does not matter appreciably. Steve senses correctly that something is wrong here, but fixes his attention on the matter of independence of the X_i. It is true that if we have prior information making them logically related, we should take this into account by a correlated prior, and this may enable us to get better estimates of α, β; but we think this is a detail, not the crucial point.

The crucial point is that if we use any fixed prior for X_i, then as the estimated line rotates we are expressing different states of knowledge about the positions of the 'true' points (X_i, Y_i) on it; and this is true even in the limit where the prior is uniform. To see this most clearly, suppose that instead of integrating out the X_i we had integrated out the Y_i with respect to independent uniform priors. A short calculation shows the surprising fact that we get a different posterior pdf for α, β: But this is just the original Gauss solution for the case that $\sigma_y = 0$; the errors are only in x_i. Merely by changing from a prior uniform in X_i to one uniform in Y_i the roles of x_i, y_i have somehow become interchanged. This is far more than we bargained for; the conventional folklore which says that prior information does not matter as long as the prior uncertainty is large compared to the width of the likelihood function, is surely true in the conventional situations one had in mind before; but now we see that the principle needs to be stated more carefully in problems with many parameters.

The reasoning here is much like that for location parameters: our prior information need not be translationally invariant, yet the likelihood function for a location parameter will have an obvious translational invariance; so a translationally invariant prior will lead to the simplest final solution. This being the case, unless we have cogent prior information which is not translationally invariant, it would be foolish not to use a uniform prior (uniform, that is, over an interval wide enough to include all the high likelihood region). Indeed, this would be actively dishonest, in effect claiming to have more information than we possess.

In our present problem, it is of course true that our prior information need not be rotationally invariant. Indeed, even if x and y are quantities of exactly the same kind, the mixture $z = x\cos\theta + y\sin\theta$ may be without any meaning.

Nevertheless, in solving this problem we shall find ourselves dealing, inevitably, with at least the mathematics of rotations in the x-y plane. You can see from the beginning that this must be so, because with Gaussian sampling distributions, only the second central moments of the data will appear in the sufficient statistic. But the collection of all those second moments forms a symmetric second rank tensor, and its minimum eigenvalue is found by reduction to diagonal form, always accomplished by a rotation of the coordinate axes. Therefore rotations will appear naturally in the likelihood function whether or not our prior information has rotational invariance. (In fact, we shall find a rotational invariance property exactly analogous to the translational invariance with a location parameter.)

This being the case, a rotationally invariant prior will lead to the (analytically) simplest solution, so unless we have cogent prior information which is not rotationally invariant, it will be prudent, both for pragmatic and philosophical reasons, to use an invariant prior.

REDUCTION OF THE RESULT

Examine the modified quadratic form that the mathematics led us to:

$$F(\alpha,\beta) \equiv \frac{Q(\alpha,\beta)}{1+\beta^2} = \frac{(s_{yy}-\beta^2 s_{xx}) + (\alpha-\hat{\alpha})^2 + 2\bar{x}(\alpha-\hat{\alpha})(\beta-\hat{\beta}) + \overline{x^2}(\beta-\hat{\beta})^2}{1+\beta^2} \tag{38}$$

It is evident by inspection that this has a single unique minimum with respect to α; by differentiation we find this is reached when $\alpha - \hat{\alpha} = -\bar{x}(\beta-\hat{\beta})$. So, keeping α constantly fixed at this value, (38) reduces to

$$F(\beta) = \frac{(s_{xx}-\beta^2 s_{xx}) + s_{xx}(\beta-\hat{\beta})^2}{1+\beta^2} \tag{39}$$

which looks quite complicated and unsymmetrical in terms of the parameter β. But β is a very unsymmetrical parameter; the real sense of the modified quadratic form appears if we set $\beta = \tan\theta$ and rewrite this in terms of the parameter θ. All the complications cancel out, and it reduces magically to the standard form

$$F(\beta) = S(\theta) = s_{xx}\sin^2\theta - 2s_{xy}\sin\theta\cos\theta + s_{yy}\cos^2\theta \tag{40}$$

which we recognize as a second rank tensor element s'_{yy} in a coordinate system (x', y') rotated an angle θ from the original one. Separating off the rotationally invariant part, this becomes

$$S(\theta) = \frac{1}{2}(s_{xx}+s_{yy}) + \frac{1}{2}(s_{xx}-s_{yy})\cos 2\theta - s_{xy}\sin 2\theta \tag{41}$$

To find the maxima and minima of this, define an angle ϕ by

$$\begin{cases} s_{xy} = R\sin\phi \\ \frac{1}{2}(s_{xx} - s_{yy}) = R\cos\phi \end{cases} \quad (-\pi < \phi \le \pi) \tag{42}$$

or,

$$\begin{aligned} \tan\phi &= \frac{2s_{xy}}{s_{xx} - s_{yy}} \\ 2R &= \sqrt{(s_{xx} - s_{yy})^2 + 4s_{xy}^2} \ge 0 \end{aligned} \tag{43}$$

Note that this defines the branch of the function so that ϕ has the same sign as s_{xy}, and if $s_{xx} > s_{yy}$, then $|\phi| < \pi/2$, otherwise $\pi/2 \le |\phi| \le \pi$. Now we have simply

$$S(\theta) = \frac{1}{2}(s_{xx} + s_{yy}) - R\cos(2\theta - \phi) \tag{44}$$

so the minimum is reached at $\theta = \theta_0 = \phi/2$, the maximum at right angles, $\theta = \theta_0 \pm \pi/2$. If the prior for α, β is rotationally invariant, we shall then estimate β as $\tan(\phi/2)$; now let us determine the accuracy of that estimate. Whatever the prior, the quasi-likelihood function, which contains all the information the data have to give us about θ, is now

$$L(\theta) = \left[\frac{1}{S(\theta)}\right]^{n/2} \tag{45}$$

But although the algebra tells us this, can we understand it intuitively in a way that makes it obvious from the start? Yes, and in fact the solution to a more general problem than the one just discussed is equally obvious, if we look at this way. Each data point has some error which may be in either x, or y, or both. But if we make the estimates $\hat{\alpha}, \hat{\beta}$, then the component of error parallel to that line contributes nothing to the error in our estimate of either α or β. Only the component of error perpendicular to the estimated line matters.

Now an error vector (e_i, f_i) has components parallel and perpendicular to a regression line of slope $\beta = \tan\theta$ of

$$e_i\cos\theta + f_i\sin\theta = \frac{e_i + \beta f_i}{\sqrt{1+\beta^2}}, \quad -e_i\sin\theta + f_i\cos\theta = \frac{-\beta e_i + f_i}{\sqrt{1+\beta^2}} \tag{46}$$

respectively. But these have mean square values of

$$\frac{\sigma_x^2 + \beta^2\sigma_y^2}{1+\beta^2}, \quad \frac{\sigma_y^2 + \beta^2\sigma_x^2}{1+\beta^2} \tag{47}$$

respectively. Therefore the quantity $\sigma^2 = \sigma_y^2 + \beta^2\sigma_x^2$ generated by our integration over the X_i was, essentially, just the mean square value of the perpendicular component of error.

Recognizing this, it is clear that we need not have considered independent sampling probabilities for e_i, f_i; whether the errors are in x, in y, or in both; and whatever the shape or orientation of the concentration ellipse, only the perpendicular component of the error can matter and we shall be led to the same result. The algebra, once we learn how to state the problem correctly, gives us this result so fast it seems like magic.

PRIOR PROBABILITIES AND TRANSFORMATION GROUPS

The transformation group principle for assigning priors in the regression problem is quite simple. Given the straight line equation $y = \alpha + \beta x$ with a prior probability element

$$f(\alpha, \beta) d\alpha d\beta \tag{48}$$

we formulate Problem P: Given a data set $D = \{(x_1, y_1) \ldots (x_n, y_n)\}$, estimate α and β. Now consider a related problem: carry out a linear coordinate transformation $(x, y) \to (x', y')$ such that in the new variables (48) takes the form

$$y' = \alpha' + \beta' x' \tag{49}$$

with a prior probability element

$$g(\alpha', \beta') d\alpha' d\beta' \tag{50}$$

and consider Problem P': Given a data set $D' = \{(x'_1, y'_1) \ldots (x'_n, y'_n)\}$, estimate α' and β'. Now if the two priors (48), (50) express the same prior information, it must be true that

$$f(\alpha, \beta) d\alpha d\beta = g(\alpha', \beta') d\alpha' d\beta' \tag{51}$$

or,

$$f(\alpha, \beta) = g(\alpha', \beta') \frac{\partial(\alpha', \beta')}{\partial(\alpha, \beta)} \tag{52}$$

This transformation equation tells how the two problems are related to each other, and it will hold whatever linear transformation we carry out, and whether or not we consider the problems P, P' to be equivalent. But now suppose that our prior information is invariant under the transformation. For example, if x is the distance from our present location to some origin of whose location we know nothing, then after the transformation x' = x+a, of walking a distance a, we are in the same state of ignorance; ignorance of one's location is a state of knowledge which is not changed by a small change in that location.

[TRUNCATED: The original draft contains multiple place-holders and ends abruptly.]

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